

The Singular Cohomology Ring of a Matroid

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Motivation: The Chow Ring of a Matroid

The **Chow ring** of a matroid M is the Chow ring of the toric variety X_{Σ_M} associated to the **Bergman fan** Σ_M .

Feichtner–Yuzvinsky Presentation of the Chow Ring [5]:

For a matroid M on $[n]$,

$$A^\bullet(X_{\Sigma_M})\mathbb{Q} \cong \frac{\mathbb{Q}[\Sigma_M]}{\left(\sum_{i \in F} x_F - \sum_{j \in G} x_G \mid i, j \in [n]\right)}$$

where $\mathbb{Q}[\Sigma_M]$ is the Stanley–Reisner ring.

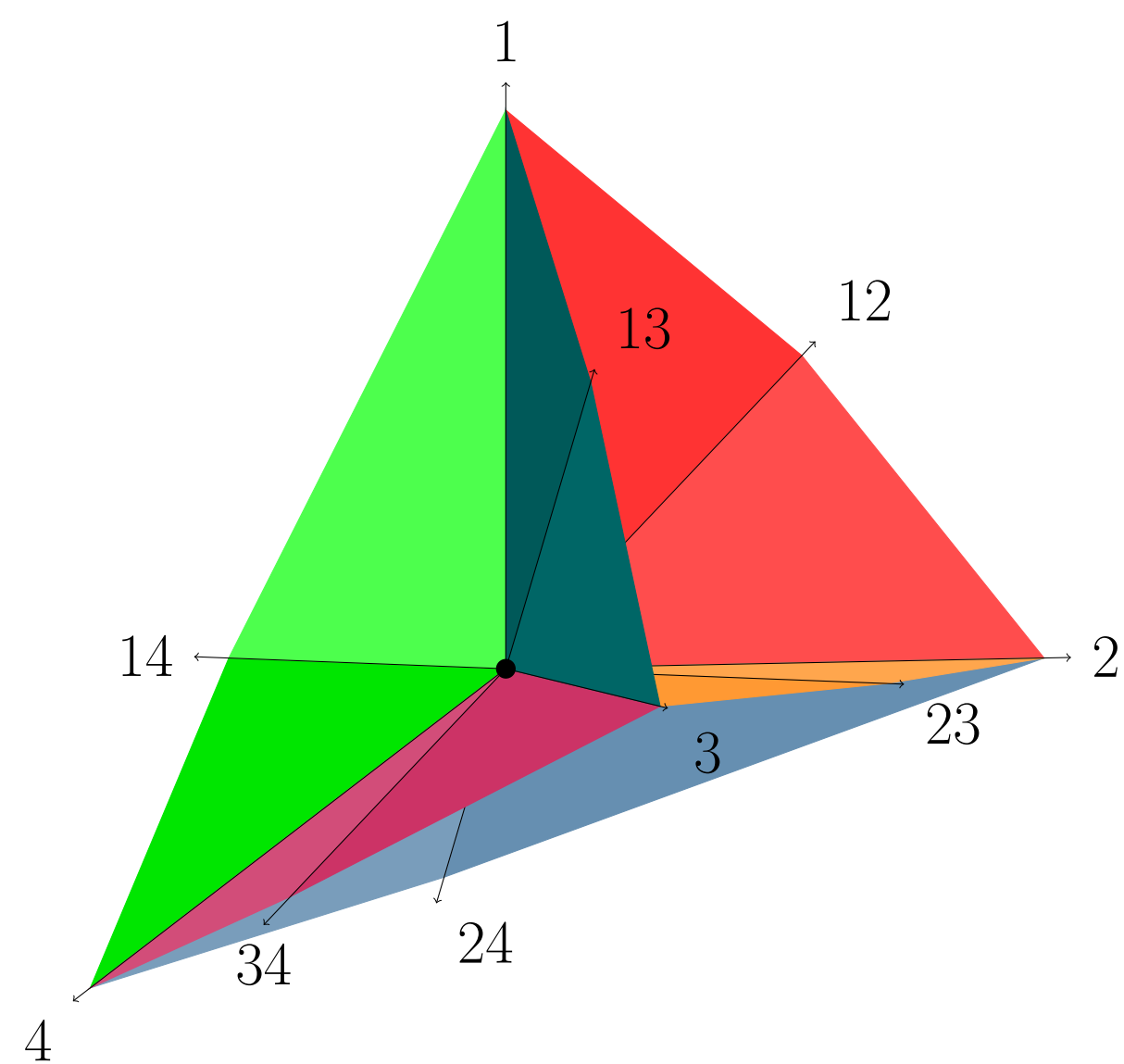


Figure 1. The Bergman fan for $U_{3,4}$.

These rings were introduced by Adiprasito, Huh, and Katz to resolve the Heron–Rota–Welsh conjecture in matroid theory [1].

Theorem [1]

The Chow ring of a matroid satisfies the **Kähler package** of:

1. Poincaré duality
2. the Hodge–Riemann relations
3. Hard Lefschetz: For $\ell \in \mathcal{K}_{\Sigma_M}$ in the *ample cone*,

$$\times \ell^{r-2i-1}: A^i(X_{\Sigma_M}) \longrightarrow A^{r-i-1}(X_{\Sigma_M})$$

is an isomorphism.

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The toric varieties X_{Σ_M} are almost always non-complete, so the Chow ring should miss some cohomology.

Questions

1. What does the **singular cohomology ring** of M , $H^\bullet(X_{\Sigma_M}, \mathbb{Q})$, look like?
2. Does it satisfy anything like the Kähler package?

Background: Cohomology of Toric Varieties

If $\Sigma \subseteq N$ is a unimodular fan, then $\mathbb{Q}[\Sigma]$ is a $\text{Sym}(N_{\mathbb{Q}}^{\vee})$ -algebra.

Franz Presentation of the Cohomology Ring [6]:

$$H^\bullet(X_\Sigma, \mathbb{Q}) \cong \text{Tor}_\bullet^{\text{Sym}(N_{\mathbb{Q}}^{\vee})}(\mathbb{Q}[\Sigma], \mathbb{Q})_\bullet.$$

The bigrading of Tor corresponds to the weight filtration:

$$\text{Gr}_{2j}^W H^{2j-i}(X_\Sigma, \mathbb{Q}) \cong \text{Tor}_i^{\text{Sym}(N_{\mathbb{Q}}^{\vee})}(\mathbb{Q}[\Sigma], \mathbb{Q})_j.$$

The cohomology can also be computed by **Koszul homology**.

Main Results

Theorem

Let M be a loopless matroid of rank r on $[n]$.

1. $\text{Gr}_j^W H^k(X_{\Sigma_M}, \mathbb{Q}) = 0$ whenever $j > n - r + k$ or $j < 2k - 2r + 2$.
2. $H^k(X_{\Sigma_M}, \mathbb{Q}) = 0$ whenever $k > n + r - 2$.

As $\text{Aut}(M)$ -representations, $\text{Gr}_{2n-2}^W H^{n+r-2}(X_{\Sigma_M}, \mathbb{Q}) \cong \text{OS}^r(M) \otimes \text{sgn}$.

We construct a combinatorial basis in Koszul homology for the singular cohomology of uniform matroids.

Example: Basis for $U_{3,4}$

	0	1
0	1	—
1	$x_4, x_{12}, x_{13}, x_{14}, x_{23}, x_{24}, x_{34}$	$x_{12} \otimes (\ell_3 - \ell_4), x_{13} \otimes (\ell_2 - \ell_4), x_{14} \otimes (\ell_2 - \ell_3), x_{23} \otimes (\ell_1 - \ell_4), x_{24} \otimes (\ell_1 - \ell_3), x_{34} \otimes (\ell_1 - \ell_2)$
2	$x_4 x_{34}$	$x_3 x_{23} \otimes (\ell_1 - \ell_4), x_4 x_{24} \otimes (\ell_1 - \ell_3), x_4 x_{34} \otimes (\ell_1 - \ell_2)$

This basis is described in terms of special chains of flats of $U_{r,n}$, and this basis gives combinatorial proofs of numerous known formulas for Chow polynomials of uniform matroids.

We use this basis to prove a Strong Lefschetz result for the singular cohomology ring.

Theorem

For generic $\ell \in \text{Gr}_2^W H^2(X_{\Sigma_{U_{r,n}}}, \mathbb{Q})$,

$$\times \ell^d: \text{Gr}_j^W H^k(X_{\Sigma_{U_{r,n}}}, \mathbb{Q}) \longrightarrow \text{Gr}_{j+2d}^W H^{k+2d}(X_{\Sigma_{U_{r,n}}}, \mathbb{Q})$$

is injective for $0 \leq d \leq r + j - 2k - 1$ and surjective for $d > r + j - 2k - 1$.

References

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